

Lab 6 Report

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DFCS 605: Digital Forensics and Cyber Investigation Foundations

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Project: Digital Forensic Investigation.

1. Summary

The goal of this lab was to practice finding process IDs (PIDs), ensuring executables are not obfuscated malware, and utilizing VirusTotal to check hashes against their database. In this lab we tested this by first finding the IP address of the virtual machine, default gateway, and DNS server. Google chrome was used to run a test a website and to find the IP address with that website, its PID, and name of executable. This PID was tracked across a variety of locations along with the same name of the executable and SHA256 hash.

2. List of Forensic Tools

The tools listed in Table 1 were used to analyze evidence.

N/A

Table 1: Forensic Tools

N/A

3. Explanation of Work

To start the lab off, I opened two windows of command prompt as administrator. In one window I ran ipconfig to get the IPv4 address of the virtual machine as seen in Figure 1. The IPv4 was 10.11.22.244/28 and a default gateway of 10.11.33.241. Next, I opened up a Google chrome browser and just typed “about:blank” to open a blank tab. After this, I ran the command “nslookup” to know the IPv4 address of the DNS server. The IP was 10.205.1.15. After typing this command a prompt is available. I typed “blankwebsite.com” which revealed the IPv4 address of that website that I typed into Google Chrome. The IP was 74.208.159.18. This can be seen in Figure 2.

Using the same command prompt I ran “netstat -ano” to look for the matching IP address, the port it is connected to, protocol, and PID. The website was connected via TCP/80 which shows that it is an unsecure connection using HTTP rather than HTTPS. Additionally, the PID was 4776. In a separate command prompt, I ran tasklist and was able to compare the chrome.exe session by the matching PID of 4776. This can be seen in Figure 3.

It was then necessary to confirm that the PID associated with the name chrome.exe was indeed chrome.exe and not an obfuscated executable. This was confirmed by finding the executable path. Once this was confirmed, I then found the SHA256 hash which is listed in section 5. I entered the same hash into VirusTotal and found that the hash was clean. This can be seen in Figure 4.

4. Results

Figure 1 shows the file signature for the file named alpha.jpg as seen in FTK Imager.

Figure 1: Results of running “ipconfig” on virtual machine

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Administrator: Command Prompt
C:\Users\Administrator>ipconfig

Windows IP Configuration

Ethernet adapter Ethernet:

    Connection-specific DNS Suffix  . : reddog.microsoft.com
    Link-local IPv6 Address . . . . . : fe80::8dbf:c830:2c4f:1239%8
    IPv4 Address. . . . . : 10.11.33.244
    Subnet Mask . . . . . : 255.255.255.240
    Default Gateway . . . . . : 10.11.33.241

Ethernet adapter Ethernet 8:

    Connection-specific DNS Suffix  . :
    Link-local IPv6 Address . . . . . : fe80::e36a:4905:9bb6:517d%5
    Autoconfiguration IPv4 Address. . : 169.254.6.31
    Subnet Mask . . . . . : 255.255.0.0
    Default Gateway . . . . . :

Ethernet adapter Ethernet 3:

    Connection-specific DNS Suffix  . :
    Link-local IPv6 Address . . . . . : fe80::2c77:1710:1814:d3b7%3
    Autoconfiguration IPv4 Address. . : 169.254.167.216
    Subnet Mask . . . . . : 255.255.0.0
    Default Gateway . . . . . :

C:\Users\Administrator>

```

Figure 2: Results of “nslookup” and “blankwebsite.com”

```

Administrator: Command Prompt - nslookup

Ethernet adapter Ethernet 8:

    Connection-specific DNS Suffix  . : 
    Link-local IPv6 Address . . . . . : fe80::e36a:4005:9bb6:517d%5
    Autoconfiguration IPv4 Address. . : 169.254.6.31
    Subnet Mask . . . . . : 255.255.0.0
    Default Gateway . . . . . : 

Ethernet adapter Ethernet 3:

    Connection-specific DNS Suffix  . : 
    Link-local IPv6 Address . . . . . : fe80::2c77:1710:1814:d3b7%3
    Autoconfiguration IPv4 Address. . : 169.254.167.216
    Subnet Mask . . . . . : 255.255.0.0
    Default Gateway . . . . . : 

C:\Users\Administrator>nslookup
Default Server: UnKnown
Address: 10.205.1.15

> blankwebsite.com
Server: UnKnown
Address: 10.205.1.15

Non-authoritative answer:
Name: blankwebsite.com
Address: 74.208.159.18
>

```

Figure 3: Results of “netstat -ano” next to “tasklist”

Administrator: Command Prompt				Administrator: Command Prompt			
TCP	0.0.0.0:22352	0.0.0.0:0	LISTENING	2900	nlshsv.exe	4276	Services
TCP	0.0.0.0:47001	0.0.0.0:0	LISTENING	4	svchost.exe	3176	Services
TCP	0.0.0.0:49664	0.0.0.0:0	LISTENING	612	svchost.exe	4448	Services
TCP	0.0.0.0:49665	0.0.0.0:0	LISTENING	1248	dllhost.exe	3996	RDP-Tcp#0
TCP	0.0.0.0:49666	0.0.0.0:0	LISTENING	1868	svchost.exe	6412	Services
TCP	0.0.0.0:49670	0.0.0.0:0	LISTENING	2592	msdtc.exe	2564	Services
TCP	0.0.0.0:49671	0.0.0.0:0	LISTENING	2944	svchost.exe	4788	Services
TCP	0.0.0.0:49673	0.0.0.0:0	LISTENING	736	svchost.exe	6500	Services
TCP	0.0.0.0:49707	0.0.0.0:0	LISTENING	728	jucheck.exe	1464	RDP-Tcp#0
TCP	10.11.33.244:139	0.0.0.0:0	LISTENING	4	MusNotifyIcon.exe	5768	RDP-Tcp#0
TCP	10.11.33.244:3389	10.13.253.15:57236	ESTABLISHED	68	svchost.exe	3816	Services
TCP	10.11.33.244:49720	168.63.129.16:32526	ESTABLISHED	3164	svchost.exe	1980	Services
TCP	10.11.33.244:49728	168.63.129.16:80	ESTABLISHED	3164	cmd.exe	4784	RDP-Tcp#0
TCP	10.11.33.244:49755	168.63.129.16:32526	ESTABLISHED	3136	conhost.exe	6784	RDP-Tcp#0
TCP	10.11.33.244:51871	192.178.155.188:5228	ESTABLISHED	4776	cmd.exe	7532	RDP-Tcp#0
TCP	10.11.33.244:54526	74.208.159.18:80	ESTABLISHED	4776	conhost.exe	3820	RDP-Tcp#0
TCP	10.11.33.244:54527	74.208.159.18:80	ESTABLISHED	4776	chrome.exe	6784	RDP-Tcp#0
TCP	10.11.33.244:54528	151.101.129.91:443	ESTABLISHED	4776	chrome.exe	5292	RDP-Tcp#0
TCP	127.0.0.1:8065	0.0.0.0:0	LISTENING	4752	chrome.exe	3216	RDP-Tcp#0
TCP	127.0.0.1:49700	127.0.0.1:49701	ESTABLISHED	3308	chrome.exe	4776	RDP-Tcp#0
TCP	127.0.0.1:49701	127.0.0.1:49700	ESTABLISHED	3308	chrome.exe	2844	RDP-Tcp#0
TCP	127.0.0.1:49721	127.0.0.1:49722	ESTABLISHED	3308	chrome.exe	5372	RDP-Tcp#0
TCP	127.0.0.1:49722	127.0.0.1:49721	ESTABLISHED	3308	chrome.exe	3884	RDP-Tcp#0
TCP	127.0.0.1:54565	127.0.0.1:8191	SYN_SENT	7908	chrome.exe	4092	RDP-Tcp#0
TCP	169.254.6.31:139	0.0.0.0:0	LISTENING	4	WmiPrvSE.exe	2292	Services
TCP	169.254.167.216:139	0.0.0.0:0	LISTENING	4	python3.exe	2208	Services
TCP	[::]:22	[::]:0	LISTENING	2892	tasklist.exe	7972	RDP-Tcp#0
TCP	[::]:135	[::]:0	LISTENING	980	WmiPrvSE.exe	7296	Services
TCP	[::]:4445	[::]:0	LISTENING	4			
TCP	[::]:4967	[::]:0	LISTENING	2864			

Figure 4: Results of the chrome.exe executable path and SHA256 hash

```

C:\Users\Administrator>wmic process where "name='chrome.exe'" get ProcessID, ExecutablePath
Node - WINDOWS
ERROR:
Description = Invalid query
{
:
:
:
C:\Users\Administrator>wmic process where "name='chrome.exe'" get ProcessID, ExecutablePath
Node - WINDOWS
ERROR:
Description = Invalid query
{
:
:
:
C:\Users\Administrator>PowerShell Get-FileHash 'C:\Program Files (x86)\Google\Chrome\Application\chrome.exe'
Algorithm Hash Path
-----
SHA256 0CAC3D17C4F4B83AD936CEAD2A7E79EFCC2E0E39EE030A84477AF95CFFC2BC84 C:\Program Files (x86)\Google\

```

Figure 5: Results of VirusTotal

VirusTotal - File - 0cac3d17c4f4b83ad936cead2a7e79efcc2e0e39ee030a84477af95cfc2bc84

0cac3d17c4f4b83ad936cead2a7e79efcc2e0e39ee030a84477af95cfc2bc84

Community Score: 56 / 72

No security vendors flagged this file as malicious

0cac3d17c4f4b83ad936cead2a7e79efcc2e0e39ee030a84477af95cfc2bc84

Size: 3.26 MB

Last Analysis Date: 15 minutes ago

peexe overlay signed 64bits

DETECTION DETAILS RELATIONS BEHAVIOR COMMUNITY 3

Join our Community and enjoy additional community insights and crowdsourced detections, plus an API key to automate checks.

Security vendors' analysis

Vendor	Result	Vendor	Result
Acronis (Static ML)	Undetected	AhnLab-V3	Undetected
Alibaba	Undetected	AliCloud	Undetected
ALYac	Undetected	Antiy-AVL	Undetected
Arcabit	Undetected	Arctic Wolf	Undetected
Avast	Undetected	AVG	Undetected
Avira (no cloud)	Undetected	Baidu	Undetected

5. Hashes of Applications

Chrome.exe SHA256:
 0CAC3D17C4F4B83AD936CEAD2A7E79EFCC2E0E39EE030A84477AF95CFFC2BC84